

3

Per-unit system, Symmetrical Components & Fault Calculations

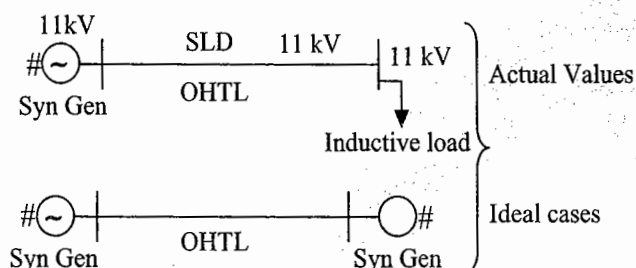
Electrical Quantities $\left\{ \begin{array}{l} \text{actual values - units} \\ \text{per unit values - No units.} \end{array} \right.$

Impedance - 20 ohms - actual.

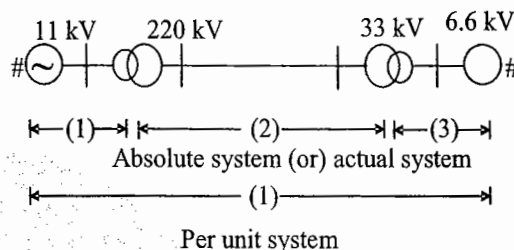
- 1.5 - P.U

Current - 100 amps - actual

- 2.0 - P. U



In above two cases can be studied electrically by using only one network so that only one network equation to be solved. Hence the taking will be less.



ACTUAL SYSTEM	PER UNIT SYSTEM
1. The given SLD is dealing with more than one operating voltage, the above SLD represented electrically with more than one n/w. So that there will be more than one n/w equation to be solved. Hence the time taking will be high	1. The given SLD is dealing with more than one operating voltage; the above SLD is represented electrically by only one n/w. So that there only n/w equation to be solved. Hence the time taking will be less.
2. In case of transformer the primary impedance is not same as secondary impedance, so it can be represented electrically by two networks.	2. per unit impedance, of transformer in primary will be same as per unit impedance of the transformer in secondary. So that it can be represented electrically by one network
$Z_p \Omega \neq Z_s \Omega$	$Z_{p.u}$

In any per unit system, the given single line diagram will be replaced by an electrical simulation networks i.e., impedance (or) reactance and admittance (or) susceptance networks in phase manner with respect to neutral point, under the assumption that 3 phase system is a balanced system, Y- Configuration and the associated electrical quantities will be expressed per unit values.



Electrical Simulation:

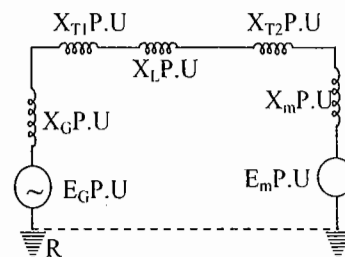
Synchronous machines → voltage source in series with impedance / reactance.

Transformer → Impedance / Reactance

Transformer line → Impedance / Reactance

Induction Motor → Impedance / Reactance

Name	Single line diagrams	Impedance/Reactance
Synchronous Generator	R << X Z = jX Rotor flux Stator flux as reference axis	
Transformer		
Transmission line		
Synchronous motor		
Induction motor		



The above simulation network is having impedance (or) reactance in per unit and also excitation for synchronous machines.

The Impedance is to be expressed in per unit in terms of the rating of Electrical equipment i.e, voltage and current or voltage and capacity in volt-ampere. (VA).

$$\text{Per unit Quantity} = \frac{\text{Actual (or) absolute quantity in the units}}{\text{Base (or) reference Quantity in the same unit}}$$

< 1.0
 > 1.0
 $= 1.0$

The magnitude of per unit Quantity depends up on base value and the selection of base value is always optional.

Alternator:

Voltage = V volts

Current = I amps

Impedance = Z ohms

$$Z_{p.u} = \frac{Z \Omega}{Z_b}$$

Choose current and voltage are base values in order to get base impedance value.

$$Z_b = \frac{V_b}{I_b} \Omega$$

$$Z_{p.u} = \frac{Z \Omega}{\frac{V_b}{I_b}} = Z \Omega \times \frac{I_b}{V_b} \text{-----(i)}$$

In AC system the rating of Electrical equipment in terms of voltage and the capacity VA.

Volts = V Volts

Capacity = VA

$$\frac{V_{Ab}}{V_b} = I_b \text{ Amps}$$



From (i) $Z_{P,U} = Z_{\Omega} \frac{VA_b}{V_b \times V_b}$

$$Z_{P,U} = Z_{\Omega} \frac{(VA)_{base}}{(V_b)^2}$$

In case of large rating of the generator,

$$Z_{PU} = Z_{\Omega} \frac{(VA)_b}{\left(\frac{V_b}{10^3}\right)^2}$$

$$Z_{PU} = Z_{\Omega} \frac{MVA_b}{(KV_b)^2} \text{----- (ii)}$$

Conversion of one per unit impedance into another per unit impedance

$$Z_{PUNEW} = Z_{pold} \frac{MVA_{new}}{MVA_{old}} \left(\frac{KV_{old}}{KV_{new}} \right)^2 \text{----- (iii)}$$

Example 3.1:

Alternator is rated as 100 MVA, 11 kV, $x = 1.2$ PU. What will be the PU reactance at 250 MVA and 13.2 kV.

Sol: $X_{PU_{new}} = 1.2 \times \frac{250}{100} \times \left(\frac{11}{13.2} \right)^2 = 2.08 \text{ P.U}$

Example 3.2:

Alternator is rated as 200 MVA, 13.2 kV, $X = 6\Omega$ and what will be the per unit reactance at 400 MVA and 18.2 kV.

Sol: $X_{PU} = 6 \times \frac{200}{(13.2)^2} = 6.88 \text{ P.U.}$

$$X_{PU_{New}} = 6.88 \times \frac{400}{200} \times \left(\frac{13.2}{18.2} \right)^2 = 7.25 \text{ per unit}$$

The per unit impedance of transformer will be same on both sides:



$$\begin{aligned} V_P & V_S \\ I_P & I_S \\ Z_P & Z_S \\ Z_{P\Omega} & \neq Z_{S\Omega} \\ Z_{PPU} & = Z_{SPU} \end{aligned}$$

$$Z_{PPU} = \frac{Z_{P\Omega}}{Z_b}$$

While calculating the primary side, use primary values are base values.

$$Z_{PPU} = \frac{Z_{P\Omega}}{Z_b} = \frac{Z_{P\Omega}}{\frac{V_b}{I_b}} = Z_{P\Omega} \frac{I_b}{V_b} = \frac{Z_{S\Omega}}{k^2} \frac{I_b}{V_b}$$

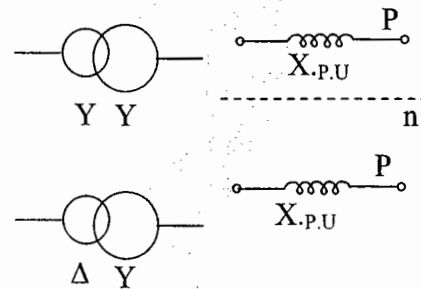
$$Z_{PPU} = \frac{Z_{S\Omega}}{\left(\frac{V_s}{V_p} \right)^2} \times \frac{I_p}{V_p} = Z_{S\Omega} \frac{V_p^2}{V_s^2} \times \frac{I_p}{V_p}$$

$$Z_{PU} = Z_{S\Omega} \frac{V_b I_b}{V_s^2} = Z_{S\Omega} \frac{V_s I_s}{V_s^2} = Z_{S\Omega} \times \frac{I_s}{V_s}$$

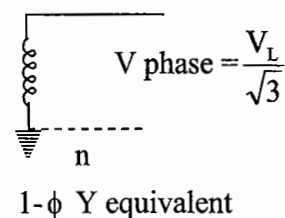
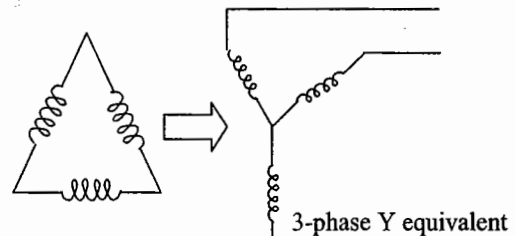
$$Z_{PU} = \frac{Z_{S\Omega}}{\frac{V_s}{I_s}} = \frac{Z_{S\Omega}}{Z_b} = Z_s \text{ PU}$$

While calculating on secondary side, secondary values are base values.

So that PU reactance in primary will be same PU reactance in secondary.



In case of ΔY T/F, reactance P.U. is same on both sides.



The Δ -connected winding will be converted into 1- ϕ . Y equivalent, the phase voltage assumed as the original line to line Δ -connected system. The P.U. reactance will be same on both sides.

Symmetrical Components

In case of balanced load, the voltage at load point will be evaluated by using per phase reactance network along with PU values. The time taking is less, because only one phase is sufficient to steady the system.

Unbalanced Condition: The occurrence of fault in 3- ϕ system will result as the 3- ϕ system will be come unbalanced. The unbalanced load voltages will be evaluated by having individual representation of phase.

There will three network equations to solve in order to evaluate the unbalanced load voltage. The time taking will be high. In order to reduce the time taking for evaluation of unbalanced load voltages, it is proposed to express any unbalanced electrical quantity by a set of 3 balanced electrical quantities and set of three balanced electrical quantities called symmetrical quantities (or) symmetrical components namely,

1. Positive sequence components, -1
2. Negative sequence components - 2
3. Zero sequence components - 0

$$V_R = V_{R1} + V_{R2} + V_{R0}$$

$$V_Y = V_{Y1} + V_{Y2} + V_{Y0}$$

$$V_B = V_{B1} + V_{B2} + V_{B0}$$

$$I_R = I_{R1} + I_{R2} + I_{R0}$$

$$I_Y = I_{Y1} + I_{Y2} + I_{Y0}$$

$$I_B = I_{B1} + I_{B2} + I_{B0}$$

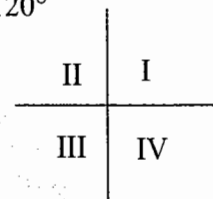
The occurrence of fault will be unbalance in phases, so that voltage and current will be express by symmetrical components. However power can not be express by symmetrical components because, power calculation will be always 3- ϕ .

Three unbalanced

$$\left. \begin{aligned} V_R &= V_{R1} + V_{R2} + V_{R0} \\ V_Y &= V_{Y1} + V_{Y2} + V_{Y0} \\ V_B &= V_{B1} + V_{B2} + V_{B0} \end{aligned} \right\} \begin{array}{l} 9 \text{ Symmetrical} \\ \text{Components} \end{array}$$

In order to reduce no. of symmetrical components, it is proposed to replace the symmetrical components of Y-phase, B-phase, in terms of R-phase by using suitable notation which is denoted by K and it is defined as unity magnitude with 120° phase displacement in anti clock wise direction.

$$K = 1 \angle 120^\circ$$



$$K^2 = K \times K = 1 \angle 120^\circ \times 1 \angle 120^\circ$$

$$= 1 \angle 240^\circ = -0.5 - j0.867$$

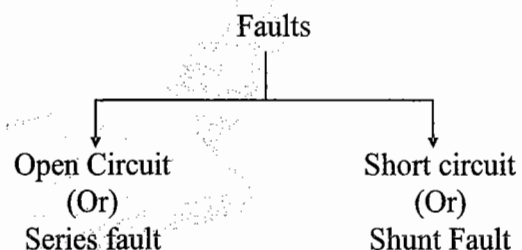
$$K^3 = K \times K \times K = 1 \angle 360^\circ = 1.0 + j0.0$$

$$K + K^2 + K^3 = 0; 1 + K + K^2 = 0$$

$$K^4 = K^3 K = K$$

$$K^5 = K^3 K^2 = K^2$$

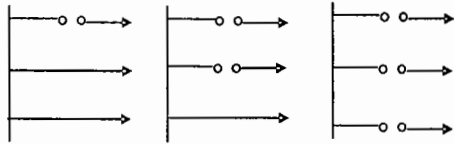
$$K^6 = K^3 K^3 = K^3 = 1$$



The open circuit faults are due to melting of the conductor because of over loading, or switching of the conductor due to wind effect or opening of the circuit breaker contact.

Unbalanced (or) unsymmetrical faults

- i. Melting of the conductor in one phase or opening of the circuit breaker in one phase
- ii. Melting of the conductor in two phases or opening of the circuit breaker in any two phases.



The open circuit occurs in all three phases the system will work at no load condition then no load voltages same as rated voltages. Hence it will not be treated as fault.

Short Circuit (or) shunt fault:

A short circuit fault will occurs due to failure of insulation or falling tree branches on the wires

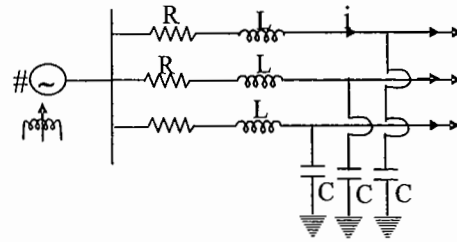
Name	Types of Faults	Diagrams	Percent age
Unbalanced or unsymmetrical	i). Line to Ground fault (LG)		70
	ii) Line to line fault (LL)		15
	iii) Line to line to ground fault (LLG)		10
Balanced or Symmetrical	i) Line to line to line fault (LLL)		5
	ii) Line to line to ground (LLLG)		

A short circuit occurs in all three phases will result as there will be short circuit current in the phase. The short circuit current is more than load currents. The failure of insulation is more common when compare to the melting of conductor, hence short circuit faults are more common when compare to open circuit.

Short circuit fault:

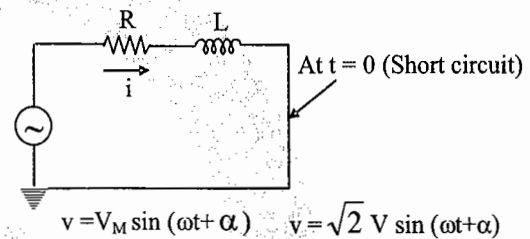
Transmission line:

The transmission line is supplied by synchronous generator and it is working at steady state condition with a balanced load. The voltages and currents are expressed in rms values.

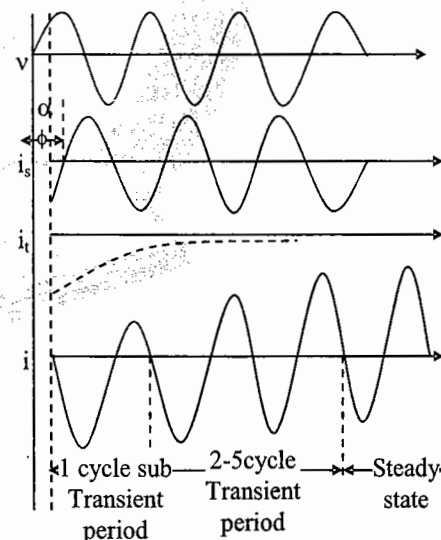


Short circuit on R-L network: (Transmission line)

At $t = 0$ the short circuit is taking place in transmission line and the capacitance is shorted. The transmission line is replaced by R-L series circuit with a constant excitation voltage of generator.



α is the angle between fault point and zero cross over point of a sinusoidal wave form.



Instantaneous current is an unsymmetrical wave form, but sinusoidal due to the presence of dc offset current

$$R_i + L \frac{di}{dt} = \sqrt{2} V \sin(\omega t + \alpha)$$

$$i = I_{\max} \sin(\omega t + \alpha - \phi) + k e^{-t/\tau}$$

$$i = \frac{V_{\max}}{2} \sin(\omega t + \alpha - \phi) + k e^{-t/\tau}$$

$$i = \frac{\sqrt{2}}{Z} V \sin(\omega t + \alpha - \phi) + k e^{-t/\tau}$$

$$i = i_s + i_t$$

i_s = symmetrical sinusoidal steady state current (ac component)

i_t = transient current (or) exponentially decaying component (DC effect) in one direction

$$\phi = \tan^{-1}\left(\frac{X}{R}\right) \text{ and } \tau = \frac{L}{R}$$

The magnitude of dc effect can be obtained by using the initial condition of the network i.e. inductance can not allow sudden change of current at $t = 0$
 $i(0) = 0$

$$0 = \frac{\sqrt{2}}{Z} V \sin(0 + \alpha - \phi) + k$$

$$k = -\frac{\sqrt{2}V}{Z} \sin(\alpha - \phi)$$

Case (i): $\alpha - \phi = 0$ (or) $\alpha = \phi$

No DC effect

$$i = i_s$$

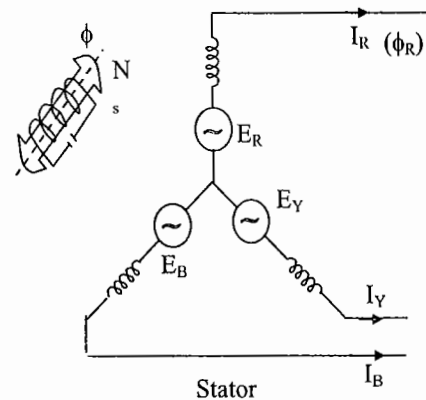
Case (ii): $\alpha - \phi = \frac{\pi}{2}$, Maximum dc effect

$$K = \frac{-\sqrt{2}V}{Z}$$

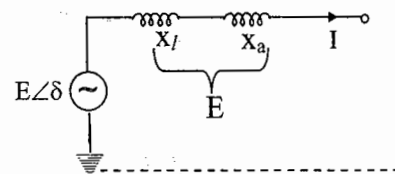
$$i = \frac{\sqrt{2}V}{Z} \sin(\omega t + \alpha - \phi) - \frac{\sqrt{2}V}{Z} \sin(\alpha - \phi) e^{-t/\tau}$$

The momentary current of the fault is having dc offset value the heat that can be produced by dc offset value is zero. So that the fault current which can be calculated is symmetrical short circuit current and it can be expressed in r.m.s. value.

Synchronous generator before fault:



Most of the loads are driven by inductive nature so that the nature of stator flux inductive the rotating flux of stator will influence the rotor flux which is called as armature reaction and it will be slightly inductive nature. It is represented by armature reactance. There is a leakage flux in the stator winding and it is represented by leakage reactance. The electrical network is shown below.



At $t = 0$, there is a short circuit fault on the terminals of generator will result as the current delivered to load is zero. But there will be current up to fault point which is called as S.C. current and it is purely inductive so that the rotor air gap flux is totally demagnetized due to armature reaction. The magnetic decoupling between rotor and stator, which can be also called as asynchronous (or) unstable.

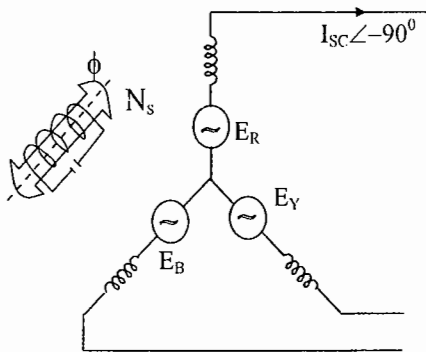
$$i = i_s + i_t$$

i_s = ac component

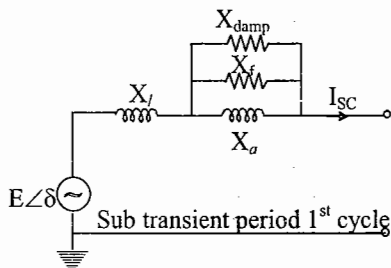
i_t = dc effect

ϕ_s = rotating at synchronous speed

ϕ_t is a static flux



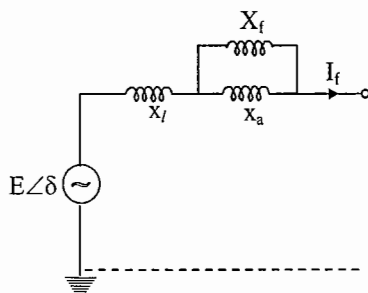
In order to maintain stability it is require to opposes armature reaction by strengthen air gap flux with the help of rotor field winding and damper winding due to flux produced by dc off set current. The electrical equivalent is as shown below to deliver the current.



Subtransient reactance

$$X_d'' = X_l + (X_a) \parallel (X_f) \parallel (X_{damp})$$

The counter emf that can be present that depends upon time constant of the elements. The damper winding having more resistance. So that time constant is less the counter emf will disappear from damper. It will be continued from to field winding. The element equivalent is as shown below



Transient period

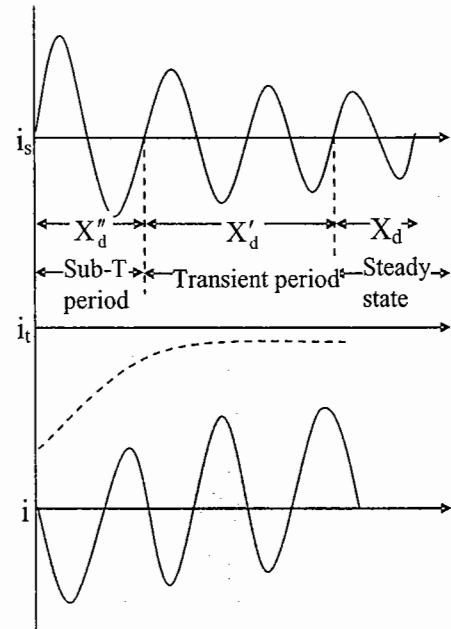
2 – 5 cycles

Transient reactance

$$X_d' = X_l + (X_a) \parallel X_f$$

The synchronous generator will brought back to steady state condition by isolating fault with help of relay and C.B.

$X_d'' < X_d' < X_d$ (synchronous generator.) The sub transient reactance will be very less compare to steady state reactance, so that sub transient currents are very high compare to steady state currents.



In case of synchronous generator the ac component is a symmetrical sinusoidal waveform with a variable current due to change reactance because of Armature reaction. However, it is measured in rms value.

Fault name	Occurrence of fault	Severity of fault (Isc)
Line to ground	Most common (or) More frequent	Less severe
Line to line		
Line to line to ground		
Line to line to line	Less common (or) Less frequent	More severe

In three phases to ground fault the potential of ground zero. It can draw more current when compare to three phase fault.